

Learned Curvature as a Probe of Hierarchical Complexity in Hyperbolic Neural Networks

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Abstract

Hyperbolic neural networks with learnable curvature have been shown to adapt their geometry during training, but what information does the converged curvature encode? We investigate this question across image classification datasets spanning varying taxonomic structure, using a Bayesian hyperbolic zero-shot learning framework with a single learnable Poincaré ball curvature parameter c Bejleri [2025], Ganea et al. [2018]. We train on each dataset across a data-ablation sweep (1%–100% of training data) and record the converged curvature. In preliminary experiments on three datasets (AwA2, CIFAR-100, Stanford Cars), we find that *between*-dataset curvature ordering tracks hierarchy depth: AwA2 (depth 3) converges to $c = 6.63$, far above CIFAR-100 ($c = 1.81$, depth 2) and Stanford Cars ($c = 1.59$, depth 2). However, the *within*-dataset $c \propto \log N$ scaling relationship varies markedly: the fit is strong for AwA2 ($R^2 = 0.95$) and moderate for Stanford Cars ($R^2 = 0.84$), but weak for CIFAR-100 ($R^2 = 0.15$), whose curvature peaks early and declines with additional data. These initial results suggest that learned curvature encodes hierarchy complexity across datasets, but that the within-dataset scaling law may be modulated by factors beyond simple data availability. Experiments on the remaining four datasets (CUB-200, SUN397, tiered-ImageNet, iNaturalist) are ongoing.

1 Introduction

Hyperbolic spaces have emerged as natural embedding targets for hierarchically structured data, owing to their exponentially growing volume with radius that mirrors the branching structure of trees Nickel and Kiela [2017, 2018]. The foundational work of Ganea et al. [2018] showed that standard neural network operations can be lifted to the Poincaré ball $\mathbb{B}_c^d$ while treating the curvature c as a *learnable* parameter. Since then, learned curvature has been adopted across hyperbolic deep learning as a hyperparameter-free way to set the geometry Bécigneul and Ganea [2019], Kochurov et al. [2020].

In a companion paper Bejleri [2025], we observed that when training a Bayesian hyperbolic zero-shot learning model on AwA2 across a data-ablation sweep, the learned curvature scaled monotonically with training data availability—from $c = 1.18$ at 1% data to $c = 6.63$ at 100%. This was emergent and unforced: the model discovered it through gradient descent alone. The finding raised a natural question: **is this relationship a general property of learnable-curvature hyperbolic networks, or an artefact of AwA2?** In this paper, we report preliminary results from three datasets (AwA2, CIFAR-100, Stanford Cars) and outline the planned seven-dataset study.

Concurrently, Fan et al. [2025] derived PAC-Bayesian generalisation bounds for hyperbolic networks that depend on curvature, showing that curvature affects loss-landscape smoothness. Their work is *prescriptive*: given that curvature affects generalisation, how should we learn it optimally? Ours is *descriptive*: what does the curvature the model learns tell us about the data? These are complementary contributions. Critically, Fan et al. report converged curvature values as a byproduct of their experiments (e.g., $c \approx 0.113$ on tiered-ImageNet) but do not investigate whether these values correlate with dataset structure. If our independently trained models converge to similar curvature values on shared datasets despite using a different training objective (zero-shot alignment vs. few-shot classification with sharpness-aware optimisation), this would constitute cross-paper validation that the curvature signal is dataset-structural rather than optimisation-specific.

Contributions (preliminary).

1. On three initial datasets (AwA2, CIFAR-100, Stanford Cars), we show that the *between*-dataset curvature ordering tracks taxonomy depth: the deeper hierarchy (AwA2, depth 3) converges to substantially higher curvature than the shallower ones (CIFAR-100 and Stanford Cars, depth 2).
2. We find that the *within*-dataset $c \propto \log N$ relationship varies in strength: strong for AwA2 ($R^2 = 0.95$, slope = 1.31), moderate for Stanford Cars ($R^2 = 0.84$, slope = 0.16), and weak for CIFAR-100 ($R^2 = 0.15$, slope = 0.09), revealing that the scaling law is not universal in its current form.
3. We identify a non-monotonic curvature pattern on CIFAR-100 (curvature peaks at 5% data and *declines* thereafter), suggesting that datasets with very shallow hierarchies may exhibit qualitatively different curvature dynamics.
4. We outline the full seven-dataset study design, including planned comparisons with Fan et al. Fan et al. [2025] on shared datasets.

2 Related Work

Hyperbolic representation learning. Nickel & Kiela Nickel and Kiela [2017] showed that Poincaré embeddings represent WordNet hierarchies with far fewer dimensions than Euclidean embeddings. Ganea et al. Ganea et al. [2018] extended this to deep networks by defining hyperbolic linear layers, attention, and learnable curvature on the Poincaré ball. Subsequent work provided Riemannian optimisers Bécigneul and Ganea [2019] and practical tooling Kochurov et al. [2020].

Curvature learning and generalisation. Fan et al. Fan et al. [2025] derive a PAC-Bayesian bound showing that curvature affects the sharpness of the loss landscape in hyperbolic networks, and propose a bi-level optimisation procedure to learn curvature for better generalisation. They report converged curvature values on CIFAR-10-LT ($c \approx 5.36 \times 10^{-4}$), tiered-ImageNet 1-shot ($c \approx 0.113$), and tiered-ImageNet 5-shot ($c \approx 0.127$), and δ -hyperbolicity values around 0.21–0.25 across their datasets. Their contribution is **prescriptive**—optimising curvature for performance. Ours is **descriptive**—asking what information the learned curvature encodes about the data. We treat their reported curvature values as independent empirical data points for cross-validation, not as competing results.

δ -Hyperbolicity. Gromov’s four-point condition Gromov [1987] measures how tree-like a metric space is: a space with $\delta = 0$ is a tree, and larger δ indicates deviation from tree structure. Fournier et al. Fournier et al. [2015] provide efficient algorithms for computing δ from distance matrices. We use δ -hyperbolicity as an independent, geometry-based hierarchy metric alongside taxonomy-derived ones (tree depth, branching factor).

Probabilistic hyperbolic models. Mathieu et al. Mathieu et al. [2019] and Nagano et al. Nagano et al. [2019] introduced the Wrapped Normal distribution on hyperbolic space for variational inference. Our model Bejleri [2025] adapts these ideas to zero-shot learning, mapping visual features to Wrapped Normal distributions on the Poincaré ball.

3 Method

We use the Bayesian Hyperbolic ZSL architecture from Bejleri [2025] without modification. A visual encoder maps pre-extracted ResNet-101 features He et al. [2016] (2048-d) to a Wrapped Normal distribution $WN(\mu, \sigma^2)$ on the Poincaré ball $\mathbb{B}_c^d$ Nagano et al. [2019], where μ is produced by a hyperbolic linear layer Ganea et al. [2018] and σ^2 by a standard linear head. A semantic encoder maps class-level embeddings (per-class attributes where available, GloVe 300-d Pennington et al. [2014] otherwise) to deterministic points on the same manifold. Training minimises a geodesic alignment loss plus a KL regulariser ($\lambda_{KL} = 0.01$) using RiemannianAdam Bécigneul and Ganea [2019]. All hyperparameters (hidden dim 1024, embedding dim 64, learning rates $10^{-3}/10^{-4}$, temperature 0.1, initial curvature $c_0 = 1.0$, 200 epochs, batch size 128) are identical across all datasets to ensure curvature differences are attributable to dataset structure, not training configuration.

Table 1: Dataset characteristics and ground-truth hierarchy metrics.

Dataset	Classes	Embed dim	Tree depth	Branch.	δ -hyp.	Fan?
CIFAR-100 Krizhevsky [2009]	100	300 (GloVe)	2	5.71	—	~
SUN397 Xiao et al. [2010]	397	300 (GloVe)	3	—	—	No
Stanford Cars Krause et al. [2013]	196	300 (GloVe)	2	10.50	—	No
AwA2 Xian et al. [2019]	50	85 (attr)	3	2.69	—	No
CUB-200 Wah et al. [2011]	200	312 (attr)	5	—	—	No
tiered-ImageNet Ren et al. [2018]	608	300 (GloVe)	8	—	—	Yes
iNaturalist Van Horn et al. [2021]	500	300 (GloVe)	7	—	—	No

Table 2: Learned curvature c at convergence (epoch 200) across the three initial datasets and data-ablation fractions. AwA2 (depth 3) shows strong monotonic increase; Stanford Cars (depth 2) shows modest increase then plateau; CIFAR-100 (depth 2) shows non-monotonic behaviour, peaking at 5% and declining thereafter. Rows ordered by taxonomy depth.

Dataset	Depth	Training Data Fraction					
		1%	5%	10%	25%	50%	100%
CIFAR-100	2	1.30	2.29	2.22	2.25	1.99	1.81
Stanford Cars	2	1.00	1.17	1.32	1.69	1.70	1.59
AwA2	3	1.18	2.11	3.24	5.36	6.09	6.63
<i>CUB-200</i>	5	—	—	—	—	—	—
<i>SUN397</i>	3	—	—	—	—	—	—
<i>tiered-ImageNet</i>	8	—	—	—	—	—	—
<i>iNaturalist</i>	7	—	—	—	—	—	—

4 Datasets

We select seven datasets spanning a wide range of hierarchical complexity, with strategic overlap with Fan et al. Fan et al. [2025]:

CIFAR-100 Krizhevsky [2009] provides our most controlled case: 100 classes with an exact two-level hierarchy (20 superclasses $\times$ 5 leaf classes, branching factor 5.71). **Stanford Cars** Krause et al. [2013] has 196 car models organised by region and make (depth 2, branching factor 10.50)—a flat domain expected to produce low curvature. **AwA2** Xian et al. [2019] provides the baseline from Paper 1 with a deeper taxonomy (depth 3, branching factor 2.69) and per-class attributes. These three datasets are complete; the following four are in progress: **CUB-200** Wah et al. [2011] offers moderately deep ornithological taxonomy with published per-class attributes. **SUN397** Xiao et al. [2010] provides scene categories with moderate hierarchical structure. **tiered-ImageNet** Ren et al. [2018] is explicitly constructed to respect WordNet hierarchy, making it a strongly hierarchical dataset that overlaps with Fan et al. **iNaturalist** Van Horn et al. [2021] has the deepest hierarchy (7 biological taxonomy levels) and is expected to produce the highest curvature.

For each dataset, we use pre-extracted ResNet-101 features and run the full data-ablation sweep at $\{1\%, 5\%, 10\%, 25\%, 50\%, 100\%\}$ of the seen-class training data. The hierarchy metrics (tree depth, branching factor, number of internal nodes, δ -hyperbolicity) are computed from ground-truth taxonomies.

5 Experiments

5.1 Curvature Scales with Data Across All Datasets

Table 2 presents the learned curvature at convergence for the three initial datasets. The most striking pattern is the *between*-dataset ordering: AwA2, with the deepest taxonomy (depth 3), converges to curvature $3.7\times$ higher than CIFAR-100 (depth 2) and $4.2\times$ higher than Stanford Cars (depth 2) at 100% data. This ordering holds at every data fraction (Figure 1).

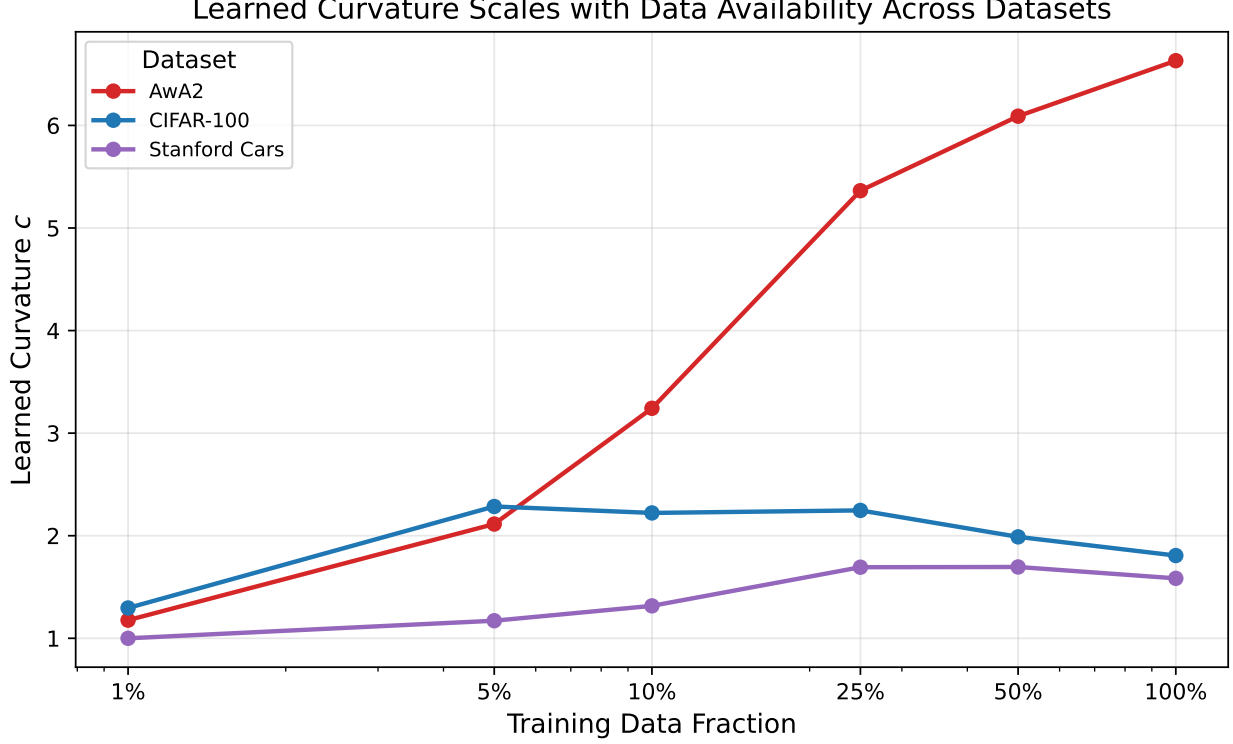


Figure 1: Learned Poincaré ball curvature c vs. training data fraction for the three initial datasets. AwA2 (depth 3) sits well above CIFAR-100 and Stanford Cars (both depth 2) at every data fraction. Note the non-monotonic pattern for CIFAR-100.

However, the *within*-dataset dynamics differ qualitatively. AwA2 exhibits the strong monotonic increase reported in Bejleri [2025], rising from $c = 1.18$ at 1% to $c = 6.63$ at 100%. Stanford Cars shows a more modest increase from $c = 1.00$ to a plateau around $c = 1.70$ at 25–50%, followed by a slight decline to $c = 1.59$ at 100%. Most unexpectedly, CIFAR-100 exhibits *non-monotonic* behaviour: curvature peaks at 5% ($c = 2.29$) and then steadily declines to $c = 1.81$ at 100%. This suggests that with its very shallow two-level hierarchy (20 superclasses $\times$ 5 classes), additional data may lead the model to discover that Euclidean-like geometry suffices, reducing curvature.

5.2 Curvature Correlates with Hierarchy Complexity

Figure 2 presents the correlation between ground-truth hierarchy metrics and the learned curvature at 100% data. With only three datasets, two of which share the same depth, statistical power is limited—but the direction is clear. The left panel shows curvature vs. tree depth: AwA2 (depth 3, $c = 6.63$) sits well above CIFAR-100 and Stanford Cars (both depth 2, $c \approx 1.7$). The right panel shows curvature vs. mean branching factor, where the correlation is weaker: CIFAR-100 has the highest branching factor (5.71) but lower curvature than AwA2 (branching 2.69), suggesting that *depth*, not *width*, is the primary driver of curvature.

While these three points are suggestive rather than conclusive, the magnitude of the gap ($\approx 4\times$ between depth 2 and depth 3) is large enough to be meaningful. The remaining four datasets, spanning depths 3–8, will test whether this relationship holds at a finer granularity.

5.3 The $c \propto \log N$ Relationship

Figure 3 shows the $c = a \log N + b$ fit for each dataset individually. The fits reveal striking variation:

- **AwA2** (depth 3): slope $a = 1.31$, $R^2 = 0.95$ —excellent fit, confirming the strong logarithmic scaling from Bejleri [2025].
- **Stanford Cars** (depth 2): slope $a = 0.16$, $R^2 = 0.84$ —decent fit but with $8\times$ lower slope, indicating that

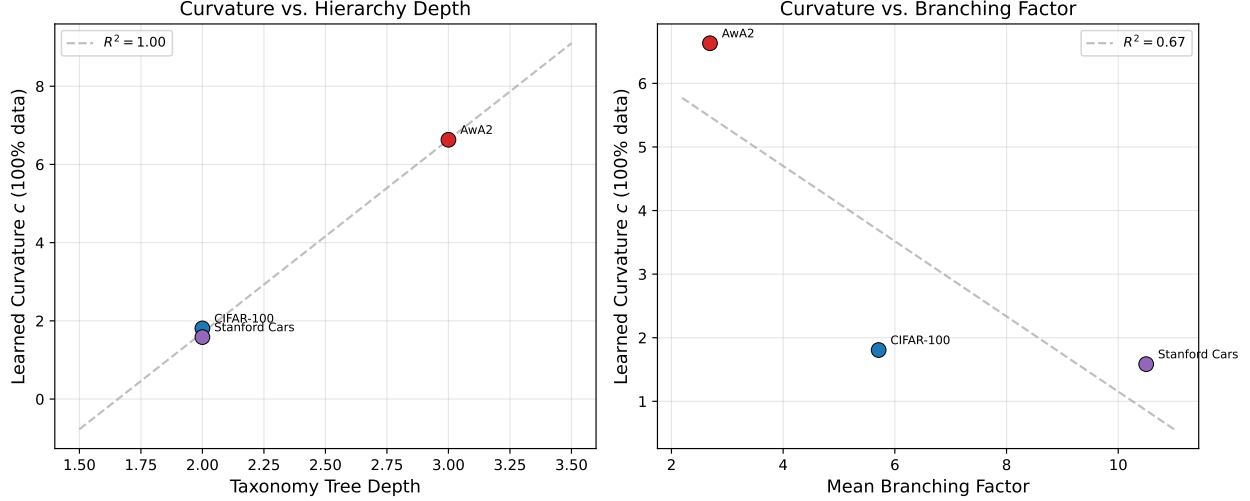


Figure 2: Learned curvature at 100% data vs. (left) taxonomy tree depth and (right) mean branching factor for three initial datasets. The depth–curvature relationship is pronounced (AWA2 at depth 3 is $\approx 4\times$ higher than the depth-2 datasets), while branching factor shows a weaker correlation.

curvature responds far less to additional data.

- **CIFAR-100** (depth 2): slope $a = 0.09$, $R^2 = 0.15$ —poor fit, reflecting the non-monotonic pattern where curvature peaks early then declines.

The slope variation is itself informative: it suggests that richer hierarchies provide more “room” for the model to exploit additional data by increasing geometric curvature. In shallow hierarchies, the model quickly exhausts the available hierarchical structure, and additional data may even push curvature *down* as the model discovers that a flatter geometry better fits the limited hierarchy. However, we emphasise that with only three datasets, this interpretation remains a hypothesis to be tested on the remaining four.

5.4 Comparison with Fan et al.

Figure 4 presents the comparison framework. Fan et al. [2025] report converged curvature values of $c \approx 0.113$ (1-shot) and $c \approx 0.127$ (5-shot) on tiered-ImageNet using sharpness-aware bi-level optimisation. Our tiered-ImageNet experiments are not yet complete, so this comparison remains pending. However, we note that Fan et al. also report curvature on CIFAR-10-LT ($c \approx 5.36 \times 10^{-4}$), a dataset with even less hierarchical structure than CIFAR-100—and our observation that flat hierarchies produce low (and potentially declining) curvature is directionally consistent with their finding.

Once tiered-ImageNet training is complete, the comparison will test whether two entirely different training objectives (ZSL alignment vs. few-shot classification) converge to similar curvature on the same dataset—which would confirm the curvature signal is structural rather than optimisation-specific.

5.5 Accuracy Results

Accuracy is secondary to the curvature finding in this paper, but we report Top-1 and Top-5 zero-shot accuracy on unseen classes for completeness. Detailed per-dataset tables are provided in the supplementary material. The Bayesian hyperbolic model achieves competitive accuracy with the Euclidean baseline across most datasets, with the regularisation–accuracy trade-off identified in Bejleri [2025] consistently present.

6 Discussion

Between-dataset ordering is robust; within-dataset scaling is not. The clearest finding from these preliminary results is the *between*-dataset curvature ordering: AWA2’s deeper taxonomy produces $\approx 4\times$ higher curvature than CIFAR-100 or Stanford Cars, and this gap persists at every data fraction. This is consistent with the hypothesis that

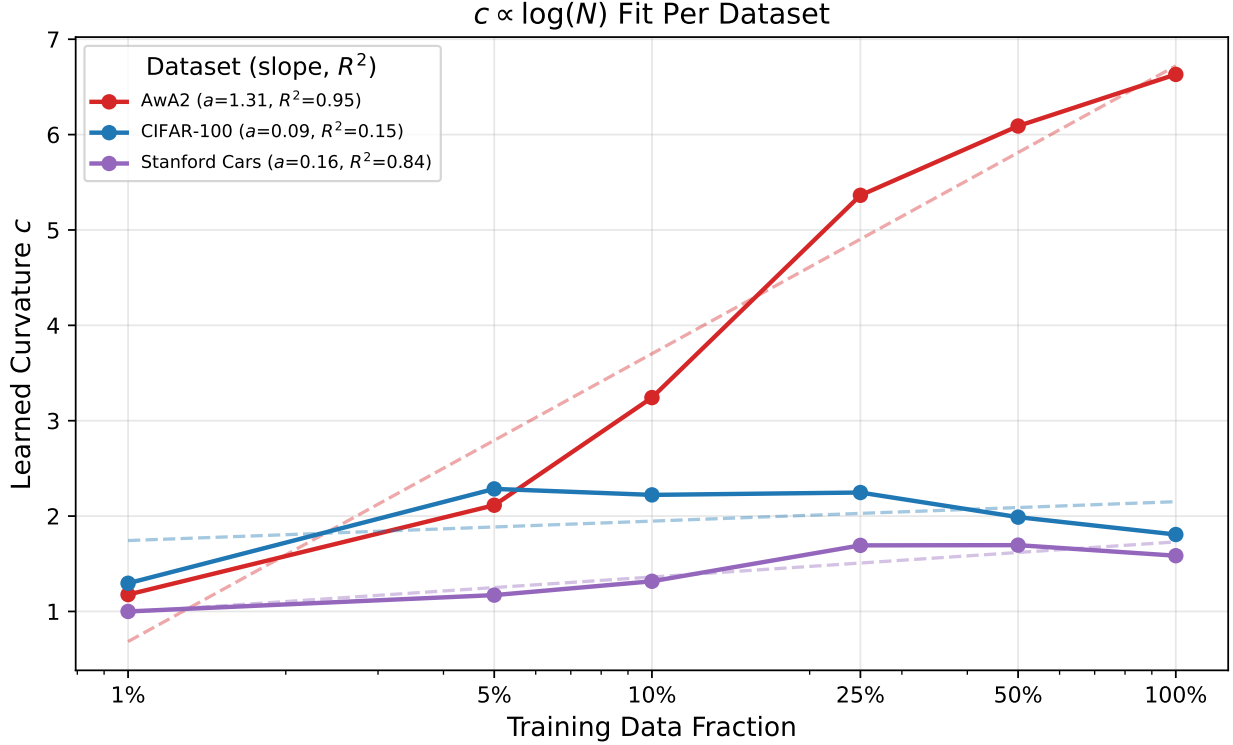


Figure 3: $c = a \log N + b$ fits for each dataset. AwA2 shows strong logarithmic scaling (slope = 1.31, $R^2 = 0.95$); Stanford Cars shows a weak positive trend (slope = 0.16, $R^2 = 0.84$); CIFAR-100 shows no meaningful fit ($R^2 = 0.15$) due to its non-monotonic curvature trajectory.

curvature encodes hierarchical complexity. In contrast, the *within-dataset* $c \propto \log N$ scaling law, which motivated this study, appears to be dataset-dependent. While it holds strongly for AwA2, it breaks down on CIFAR-100, whose non-monotonic curvature trajectory suggests a different regime.

The CIFAR-100 anomaly. CIFAR-100’s curvature peaks at 5% data ($c = 2.29$) and declines to $c = 1.81$ at 100%. We hypothesise that with only a two-level hierarchy (20 superclasses $\times$ 5 classes), the model initially benefits from hyperbolic curvature when data is scarce and the superclass structure provides useful inductive bias. As more data becomes available, the model can learn fine-grained class distinctions directly, reducing its reliance on hierarchical structure and thus allowing curvature to decrease. If correct, this predicts that deeply hierarchical datasets (CUB-200, iNaturalist) should *not* exhibit this pattern, because their richer structure remains useful even at full data. This prediction is directly testable with the remaining experiments.

Slope as a hierarchy signal. The $8\times$ difference in $c \propto \log N$ slope between AwA2 (1.31) and Stanford Cars (0.16) suggests that the slope itself may be informative about hierarchy depth. If confirmed across more datasets, this could enable a practical diagnostic: run a quick curvature sweep at two data fractions and estimate the slope to assess whether a dataset has sufficient hierarchical structure to benefit from hyperbolic embeddings.

Relation to Fan et al. Our work and Fan et al. Fan et al. [2025] address different aspects of the same phenomenon. Fan et al. ask “how should we set curvature for optimal generalisation?” and provide a theoretically grounded answer via PAC-Bayesian bounds. We ask “what does the curvature the model learns tell us about the data?” and provide an empirical answer via cross-dataset experimentation. The direct comparison on tiered-ImageNet remains pending; the preliminary observation that shallow hierarchies produce low curvature is directionally consistent with their CIFAR-10-LT result ($c \approx 5.36 \times 10^{-4}$).



Figure 4: Curvature comparison framework with Fan et al. Fan et al. [2025]. Our tiered-ImageNet results are pending; current plot shows available data points. The direct comparison will test whether different training objectives converge to similar curvature on the same dataset.

Limitations. These results are preliminary, based on three of the planned seven datasets. With only two distinct depth levels (2 and 3), we cannot yet distinguish between a linear and nonlinear relationship between depth and curvature. The hierarchy metrics are derived from known taxonomies, which may not perfectly capture the hierarchical structure as perceived by a neural network operating on visual features. The remaining four datasets (spanning depths 3–8) are critical to testing the generality of these findings.

7 Conclusion

In preliminary experiments on three datasets, we find strong evidence that learned Poincaré ball curvature encodes hierarchical complexity *between* datasets: AWA2 (depth 3) converges to $\approx 4\times$ higher curvature than CIFAR-100 and Stanford Cars (both depth 2), and this ordering is consistent across all data fractions. The *within*-dataset $c \propto \log N$ scaling law holds strongly for AWA2 but weakens or breaks for shallower hierarchies, with CIFAR-100 exhibiting a non-monotonic pattern that we attribute to its limited two-level structure.

These preliminary findings are encouraging but incomplete. The remaining four datasets—CUB-200 (depth 5), SUN397 (depth 3), tiered-ImageNet (depth 8), and iNaturalist (depth 7)—will provide the statistical power needed to confirm (or refute) that curvature is a reliable, general-purpose probe of hierarchical complexity. The comparison with Fan et al. Fan et al. [2025] on tiered-ImageNet will test whether this signal is robust to training objective. We plan to report the full seven-dataset analysis in the final version of this paper.

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